The Expanding Unknown:
A Generative Account of Curiosity

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Abstract

An explanation can settle a question while giving its recipient several new questions to ask. Continued inquiry may then reflect a change in available opportunities rather than a stronger desire for information. This paper develops that distinction through a conceptual synthesis of research on information gaps, learning progress, question generation and information seeking. Opportunity generation is defined relative to a learner's ability to formulate unresolved contrasts and identify possible discriminating observations. A conditional branching model shows how investigations can expand under a stationary pursuit rule, while making explicit the assumptions that prevent this result from becoming a general law of curiosity. A prospective experiment separates reusable explanations, supplied question menus and answer-access costs, with predictions that could favor or weaken an opportunity-based account. Existing theories already accommodate generative learning; the contribution is a clearer account of what must be measured before increased exploration is attributed to increased motivation. No new behavioral data or simulation results are reported. The argument also limits what can be inferred from pattern-guided machines, bacterial information use and pedagogical demonstrations. For education, it directs attention to the investigations a learner can formulate after an answer has been given, alongside mastery of the answer itself.

A repair that leaves questions

Imagine two people trying to understand why a bicycle chain keeps slipping. One is shown the adjustment that fixes it. The other learns enough about the mechanism to understand what the adjustment changes. Both can make the repair. The second person might now wonder whether the same fault would occur under a different load, how it would behave in another gear, or what distinguishes it from a worn component. These possibilities were not necessarily questions waiting in that person's mind before the explanation. Learning helped make them available.

Suppose the second person keeps investigating after the repair is complete. It would be reasonable to say that the explanation encouraged curiosity. It would be premature to conclude that it increased the strength of a desire to know. The explanation might have made investigation easier, revealed more useful tests, or supplied questions the person could not previously formulate. The repair is hypothetical; the ambiguity is the problem this paper addresses.

There is a familiar way to describe inquiry as the reduction of ignorance. A person lacks an answer, obtains it, and knows more. This description works for the question just settled. It becomes incomplete when that answer changes what the person can ask next. Someone can finish an investigation with less uncertainty about its original object and a longer, better specified list of things to investigate. Nothing has gone wrong with learning. The list has changed.

The argument here concerns that change. Learning can sustain inquiry by generating recognizable opportunities for further investigation, without requiring an increase in the desire to pursue each comparable opportunity. We call this *opportunity generation*. More questions can also overwhelm a learner, divert a fixed amount of attention, or reveal that further investigation would be pointless.

This is a conceptual account with a conditional model and a prospective experimental design. It does not establish that opportunity generation usually dominates other explanations of curiosity. Nor does it introduce the idea that knowledge gives rise to further questions. That idea has a substantial history within the theories considered below. The task is to make it difficult to confuse a change in what can be investigated with a change in how strongly someone wants to investigate it.

What becomes available

The relevant unknown is smaller than everything a person does not know. It consists of questions the person can currently recognize as questions: distinctions they understand, whose answers they do not already possess. Some of these questions have an imaginable route to an answer. Others remain intelligible but practically inaccessible. Learning can change either condition.

For the purposes of this account, an *investigable question* is an unresolved contrast that a learner can represent, together with a candidate observation or action that could discriminate between its alternatives under specified resources. The learner need not know a successful procedure or assign a high probability to success. An attempted experiment may fail. The requirement excludes merely repeating an unfamiliar question whose possible answers and relevance the speaker cannot explain. It also excludes treating willingness to investigate as part of the definition. Otherwise every increase in willingness would automatically count as a new opportunity.

This definition is deliberately narrower than curiosity. A person can want to know something without knowing how to find out. A scientist can ask a worthwhile question before suitable instruments exist. Such cases belong in a wider account of inquiry, but distinguishing them from presently investigable questions is useful when explaining behavior. The same resources must be specified before and after learning; giving someone a new instrument is a different intervention from teaching them to see another use for an instrument already present.

The bicycle example contains several possible changes. Learning how components interact might make a previously unformulated distinction recognizable. A demonstration might lower the effort needed to test a distinction already understood. A satisfying explanation might make another investigation feel more attractive. These changes often accompany one another. An analysis needs to keep track of each even when they occur together.

Let Q_t denote a learner's finite set of investigable questions at a particular point in a task. Taking that snapshot as given, let $I_t(q)$ indicate whether the learner initiates an investigation of $q \in Q_t$ during a fixed observation window. Write $\pi_t(q) = \Pr[I_t(q) = 1]$, conditional on the same starting state. If N_t counts the distinct questions from this initial set that are pursued, then

$$\mathbb{E}[N_t] = \sum_{q \in Q_t} \pi_t(q).$$

This is an accounting identity, not a psychological law. It requires no independence between decisions. A probability can reflect desire, expected usefulness, anticipated success, effort, social expectations and competition with other activities. The expression makes the identification problem explicit: the same observed increase in investigations can arise from a larger candidate set, different pursuit probabilities, or both.

If a newly understood explanation adds questions while the old questions retain their pursuit probabilities, expected activity increases whenever at least one added question has a positive pursuit probability. That is a conditional sufficiency claim. In a person with a full afternoon and several competing commitments, the old probabilities may fall as new questions become available. Opportunity generation might then change the content or sequence of inquiry without increasing its total duration. Counting hours would miss that change; counting clicks might mistake repeated checking for it.

There is no reason to equate $\pi_t(q)$ with the strength of curiosity. Liquin's effort experiments provide a useful empirical warning: the costs of obtaining answers affected search decisions more consistently than self-reported curiosity. The study also reports an exception in a high-confidence condition and acknowledges the limits of introspection (Liquin 2026, General Discussion). Neither search nor self-report supplies an infallible reading of a motive. Jach and colleagues likewise find only partial correspondence between personality measures and task-based information demand (Jach et al. 2024). A universal curiosity score cannot simply be assumed across these uses.

Counting questions introduces its own difficulty. "Would another gear slip?" and "Does slipping depend on the selected gear?" may name the same investigation. Conversely, one broad question can contain several distinct causal contrasts. For a bounded experiment, questions can be grouped by the alternatives they discriminate and the observations that would discriminate them. Alternative procedures for resolving the same contrast would be logged as tests of one question. The grouping rule must be fixed before testing the prediction and checked by independent coders. Outside such settings there may be no defensible single count. A change in the structure of questions can matter even when its size cannot be summarized by a scalar.

Failure to mention a question is also weak evidence that the learner could not formulate it. Recall, attention and confidence affect what gets expressed. Opportunity generation is therefore used here for a change in accessible candidates under specified conditions, not as proof that a concept has been created for the first time. Distinguishing new composition from retrieval of a previously available question would require additional probes. Even a careful elicitation task gives a sample of available questions rather than a complete inventory of the mind.

Nor does the expanding unknown imply an increase in ignorance measured in bits. An uncertainty measure refers to specified variables, alternatives and probabilities. Learning can reduce uncertainty about those variables while making another variable worth representing. Comparing the entropy of the first description with that of a richer description would confound a change in belief with a change in what is described. The present claim concerns the availability of questions. It does not require a global measure of how much remains unknown.

How an answer can sustain inquiry

Opportunity generation becomes more consequential when it recurs. An answer supplies a distinction; that distinction permits a test; the test reveals a further distinction. A person need not become progressively more eager for this sequence to continue. There must already be some disposition to pursue questions, but its intensity need not rise at every step.

A simple branching construction isolates this possibility. Let Z_n be the number of investigations begun in generation n. Assume each investigation independently reaches an answer with probability s. An answered investigation produces a random number of genuinely new, investigable questions with finite mean k, and each new question is independently pursued with probability p. Here k is conditional on reaching an answer. Failed investigations produce no children in this construction. Questions do not duplicate one another or converge on questions generated elsewhere; time and other resources do not impose an additional cap. The parameters remain constant.

These are strong assumptions, chosen to expose a possibility rather than approximate a particular population. Under them, each begun investigation has expected offspring skp. Thus, for a fixed initial Z_0,

$$\mathbb{E}[Z_{n+1} \mid Z_n] = skpZ_n, \qquad \mathbb{E}[Z_n] = Z_0(skp)^n.$$

The second equality follows by taking expectations of the first and applying it repeatedly. With s and p held fixed, changing k changes the expected course of inquiry. If $skp > 1$, expected generation size grows; if $skp < 1$, it declines. Growth in expectation does not guarantee that any particular sequence continues. The model also does not say that a larger p uniquely measures stronger desire: even that parameter can combine motivation, cost and perceived usefulness.

What the construction establishes is limited but pertinent. An expanding sequence of investigations is possible under a stationary pursuit rule when answers generate enough further opportunities. Simulating the recurrence would illustrate the assumptions, not provide independent evidence for them. No simulation or fitted parameter estimate is reported here.

Real inquiry soon departs from the construction. Two answers can disclose the same question. An apparently productive branch can lead to an untestable problem. An unanswered question can generate useful alternatives, contrary to the model's assumption about failure. A learner may abandon a whole domain after discovering that its premises were mistaken. In a finite task, available questions eventually run out. With a fixed daily budget, additional candidates compete for time instead of reproducing independently. Calling the threshold a universal law of curiosity would conceal these departures.

The construction is still useful as a counterexample to a common inference. Persistent inquiry need not imply an escalating motive, just as an expanding list of possible repairs need not imply an expanding appetite for repair. To explain an actual episode, one must determine whether the learner's questions changed, whether pursuit became less costly, whether information became

more valuable, and whether willingness changed on comparable opportunities. Persistence alone decides none of these matters.

What earlier accounts already explain

The information-gap account is an obvious predecessor. Loewenstein makes curiosity depend on a perceived gap between present knowledge and an informational reference point. Crucially, the reference point is not fixed. He explicitly describes new information as changing a person's conception of what there is to know (Loewenstein 1994, 87–90). The generative argument can therefore be read as a specification of a process already allowed by information-gap theory. Its extra demand is to investigate how a question becomes representable before interpreting changes in the wish to answer it.

Learning-progress accounts provide another route to continuing exploration. Schmidhuber's framework assigns intrinsic reward to improvements in a predictive or compressive model (Schmidhuber 2010). Oudeyer, Kaplan and Hafner use learning progress to organize autonomous exploration and sequences of increasing task difficulty in robots (Oudeyer et al. 2007). In human experiments, Ten and colleagues find that models combining competence and learning progress account for activity selection better than the alternatives they tested (Ten et al. 2021). These approaches already explain why a learner might move on after mastering a task rather than stop learning altogether. They should not be redescribed as theories of passively filling a finite container.

The distinction lies in what an explanation is asked to track. Two learners may value progress similarly while differing in which experiments they can formulate. If a model updates values over a supplied menu, it can explain selection from that menu without yet explaining how the menu was generated. A progress model that also generates candidates would make the two accounts compatible.

Patankar and colleagues go further in modeling inquiry through growing knowledge networks, including their increasing flexibility (Patankar et al. 2023). This is a close conceptual predecessor, since the knowledge representation itself changes as inquiry proceeds. Liquin, Rhodes and Gureckis provide experimental evidence that children and adults reuse prior questions and recombine their components; reuse is sensitive to a question's current informativeness. Their evidence for recombination is weaker under one of the structural measures than under semantic similarity (Liquin et al. 2025, Study 2 Discussion). Prior questions can help generate later candidates, although that result does not establish every proposed route from an explanation to a new question.

The strongest rival to an opportunity-based explanation of increased exploration is that the same lesson changes the value of further information. Dubey and Griffiths develop a rational account based on the expected usefulness of knowledge (Dubey and Griffiths 2020). An explanation of the bicycle mechanism might reveal that a test would prevent a costly failure. Further search would then be intelligible even if the learner had understood that test all along. It would be a mistake

to count the newly attractive test as newly representable merely because the person had not previously mentioned it.

Motivational reinforcement is also a substantive alternative. Abir and colleagues find that satisfying answers are associated with increased subsequent seeking for semantically related information, and interpret the pattern in terms of a learned information-seeking policy. Their discussion notes limitations in mapping satisfaction and expectation measures onto reinforcement-learning quantities (Abir et al. 2026). The finding is consistent with learning changing willingness; it is not a reason to infer that willingness changed whenever inquiry continues. In particular, answer satisfaction was not an independently randomized motivational variable that could settle the present identification problem.

These accounts need not compete for exclusive ownership of a learning episode. A reusable explanation can produce a candidate question, make its answer look useful and reduce the effort of testing it. The narrower empirical issue is whether candidate generation contributes anything once these other changes are considered. If it does not, the proposed distinction may remain conceptually clear without explaining the behavior in that setting. A study must allow that outcome.

A study that could distinguish the mechanisms

Consider a task involving unfamiliar mechanical devices represented on a screen. Participants can alter inputs and observe outputs. Each device belongs to a family with shared causal structure, but differs in some connections. The task must allow participants to express alternative mechanisms and propose interventions that distinguish them. Its purpose is to measure which investigations become available after learning.

Participants would first receive common training in the interface and the elementary operations. They would then be randomly assigned either a local explanation resolving one device's behavior or an explanation that also identifies a reusable relation within the device family. Both explanations must correctly settle the initial question. Their length, readability and presentation should be matched as far as possible, and comprehension checked. The contrast intentionally changes what can transfer. Matching the number of words does not make the explanations informationally equivalent, and the study should not pretend otherwise.

The next phase would present a new device from the same family without offering a menu of questions. Participants would be asked what remains uncertain, what alternative outcomes they envisage, and what they could do to distinguish them. A separate exploration period would allow, but not require, actual investigation. Question elicitation and exploration should also be tested in separate groups: asking people to list questions may itself generate the behavior being explained. The design therefore needs a no-elicitation condition rather than treating elicitation as a neutral measurement.

Before collecting the main data, the researchers would specify how candidate questions are individuated. Independent coders, unaware of condition, would classify the contrast and proposed test, identify duplicates, and assess whether an answer could change the participant's account of

the device. Reliability should be established on pilot material. Parallel task forms would help assess baseline candidate availability without repeating the exact questions later counted as additions. Verbal fluency must not become the hidden outcome: participants could supplement words with selected components, sketches or enacted tests. A short description with a discriminating intervention can express a question more clearly than a paragraph of technical vocabulary.

Question count would be reported alongside question type, redundancy, testability and subsequent use. A long list of interchangeable or unanswerable questions would provide little support for the account. The important prediction is a change in the set of distinct, meaningful investigations the participant can formulate, followed by exploration that draws on those additions. The resulting estimates would remain specific to this task and coding scheme.

There should also be a way to intervene on candidate availability. In a parallel, independently randomized condition, participants would receive a common menu of possible investigations, with the contrasts explained well enough that comprehension can be checked. If the reusable explanation mainly helps people generate candidates, part of its advantage in selecting those investigations should diminish when candidates are supplied to everyone. Comparison with a self-generated condition would make this prediction more specific than a general expectation of "more exploration."

The common menu is not a perfect control. It can teach, attract attention, reduce planning effort or imply that the experimenter values particular questions. Providing the same words also does not ensure the same understanding. Menu trials should therefore be analyzed as an intervention with several possible effects, supported by comprehension and usefulness measures. A reduced explanation effect under the menu would be evidence consistent with a candidate-generation bottleneck. It would not, by itself, identify a unique psychological mediator.

Costs should be varied separately, after the initial explanations and candidate measures, for example by assigning different delays to otherwise identical opportunities to observe an outcome. This manipulation concerns answer access, not the cognitive effort of understanding the device. It can test whether search changes when a known practical cost changes, while leaving candidate generation earlier in the procedure untouched. Additional measures would ask about anticipated success, expected usefulness, satisfaction with the first explanation, and curiosity about comparable specified questions. Ratings should be timed so that asking them does not silently become another intervention in every condition.

The analysis would begin with the randomized contrasts and their uncertainty. The experiment would not first select participants who passed a post-lesson competence threshold, or regress away all measured competence, and then call the remainder a pure generative effect. The explanation can change competence; conditioning on that consequence changes the estimand and can introduce selection problems. If prior ability also affects both competence and exploration, selecting successful learners can make the treatment groups differ in prior ability despite random assignment. Common pretraining, baseline measures and separate assessments of what participants understood are preferable to treating every post-treatment difference as a nuisance to remove. Mediation claims would require further assumptions and a design capable of supporting them.

Evidence favoring opportunity generation would include more distinct investigable contrasts after the reusable explanation, exploration directed toward those contrasts, and attenuation of the exploration difference when an understood common menu supplies them. A corresponding increase in rated desire would suggest a mixed mechanism rather than invalidate candidate generation. To argue that desire remained practically unchanged, the study would need a prespecified meaningful equivalence range, sufficiently precise estimates, and measures sensitive enough to detect the relevant change. A nonsignificant rating difference would not suffice.

Several outcomes would weaken the account. The explanation might improve answers to supplied questions but produce no new candidate investigations. Participants might list more questions only when instructed to do so, with no change in unprompted exploration. The effect could consist entirely of paraphrases. Or increased search might persist under an understood common menu and closely track revised usefulness, with no evidence of additional candidates. Such results would limit the proposed explanation in this domain. They would also help distinguish a failure of the manipulation or measurement from a failure of the mechanism itself. None of these comparisons has been conducted for this paper; they specify what remains to be learned.

Cheap information and the desire to obtain it

The distinction between having an opportunity and being moved to pursue it also limits what can be inferred from artificial or biological information processing. A useful representation can let an existing procedure accomplish more with the same resources. That advantage does not settle why an agent initiates inquiry, or whether it experiences anything resembling human curiosity.

Ertle, Levin and Scheutz investigate pattern-guided maze exploration, using fractal images and artworks as inputs or sources of action sequences. Structured patterns improve exploration over shuffled counterparts in their simple-mapping comparisons. In their learning-agent comparisons, replacing fractal input with noise does not significantly change certain reported effects (Ertle et al. 2025, secs. V.A, V.C–D). The latter result is not proof of equivalence, but it limits an inference that a distinctive mathematical structure explains every advantage. Across these conditions, the mapping, environment, training objective and computational resources are part of the explanation.

A "free lunch" in this setting concerns a benefit relative to a particular accounting of design or learning costs. It does not show that a system's wish to explore has appeared without antecedents. In a reward-driven architecture, the objective that makes progress valuable must be specified. A passive compressor supplies a simple counterexample: it can improve its representation of incoming observations while having no mechanism for choosing another observation. Compression gains alone are therefore insufficient to explain active inquiry. An active system must additionally connect what it learns to the selection of further observations.

Biological examples require equal care. Mattingly and colleagues relate information acquired by the chemotaxis pathway of *Escherichia coli* to performance in climbing chemical gradients (Mattingly et al. 2021). The study makes information use measurable in a living system without establishing a desire for information as such. Navigation, adaptive regulation, candidate generation and

felt curiosity are different explanatory targets. Calling them all curiosity would remove precisely the distinctions needed to investigate their relations.

The proposed human account assumes a learner capable of representing alternatives and some disposition to investigate them. It explains a possible continuation of inquiry under those conditions. It offers no derivation of that disposition from matter, information or mathematical possibility. Extending the account to other organisms or machines would require specifying what plays the role of a question, what counts as an answer, and what selects further action. The definitions would have to earn their use through explanatory work.

What a lesson leaves open

An explanation arrives from outside a learner, yet can change what that learner wants to know. There is no contradiction in this. Calling a motive intrinsic distinguishes the value of learning from a separate reward; it does not make the motive causally insulated from encounters, language or other people. Conversely, inducing the outward behavior of inquiry does not guarantee a corresponding desire. A person can ask the required question to finish an assignment.

Pedagogy makes the distinction practically important. In Bonawitz and colleagues' toy experiments, children explored less broadly after certain pedagogical demonstrations than under comparison conditions. The authors explain this in terms of what a learner can infer from an informative teacher's choice to demonstrate one function and omit others (Bonawitz et al. 2011). This result concerns a particular inference in a particular task. It cannot establish that teaching invariably suppresses curiosity or that withholding instruction generally helps learners.

The two possibilities need to be considered together. Instruction can supply the concepts with which a learner recognizes another question. It can also suggest that the demonstrated answer exhausts what is worth knowing. An institution interested in independent inquiry should therefore attend to what its lessons make askable and what their presentation implies has already been settled. This is an educational judgment defended by the account.

One practical consequence would be to assess a lesson partly through the learner's next investigation. Can the learner identify a limitation, vary a condition, propose an alternative explanation, or say what observation would change a conclusion? Those abilities cannot be read directly from the number of questions asked. They also require time and permission to pursue an answer that may fall outside the scheduled sequence. Teaching a reusable explanation while penalizing every departure from the planned task could give a learner opportunities that the institution then makes costly to use.

There are reasons to close inquiry as well as to extend it. A novice needs reliable answers; some procedures must be learned accurately; limited time forces choices. The argument supplies no warrant for endless questioning or for treating successful completion as intellectual failure. It asks for a distinction between an answer that resolves a question and a setting that prevents any further question from counting.

Return to the repaired bicycle. The useful outcome may be a machine that works and an owner who can get on with the day. It may also be a person who now understands why another test is worth making. What has changed in the second case need not be the person's appetite for knowledge. An explanation has given the next investigation an intelligible form. To judge whether learning has sustained inquiry, we have to look beyond the answer that was supplied and ask what the learner can now ask without us.

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